Python Code for All Four Data Visualizations

Shortened versions of the Finance, Healthcare, Census, and Geospatial programs from your uploaded PDF. Healthcare code uses the separately provided healthcare_dataset.csv file.

Required Libraries

```
pip install pandas numpy matplotlib seaborn geopandas folium
```

1. Finance — Stock Price Visualization

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
df = pd.DataFrame({
    'Date': pd.date_range('2024-01-01', periods=30),
    'StockPrice': np.random.randint(100, 200, 30)
})
df.to_csv('finance_dataset.csv', index=False)

df['Date'] = pd.to_datetime(df['Date'])
plt.plot(df['Date'], df['StockPrice'], 'bo--')
plt.title('Stock Prices Over Time')
plt.xlabel('Date')
plt.ylabel('Stock Price')
plt.grid()
plt.show()
```

2. Healthcare — Data Analysis and Visualization

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = pd.read_csv('healthcare_dataset.csv')
print(df.info())
print(df.head())
print(df.describe())
print(df.isnull().sum())

sns.countplot(x='Gender', data=df)
plt.title('Patient Gender Distribution')
plt.show()

sns.boxplot(x='Gender', y='Age', data=df)
plt.title('Age Distribution by Gender')
plt.show()

sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

3. Census — Population by Age Group

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    'Age': ['0-4', '5-9', '10-14', '15-19', '20-24', '25-29'],
    'Population': [120000, 150000, 180000, 200000, 250000, 220000]
})

df.plot(x='Age', y='Population', kind='bar',
        color='skyblue', legend=False)
plt.title('Population Distribution by Age Group')
plt.xlabel('Age Group')
plt.ylabel('Population')
plt.show()
```

4. Geospatial — Map Visualization

```
# Install first: pip install geopandas folium
import geopandas as gpd
import folium
import matplotlib.pyplot as plt

df = gpd.GeoDataFrame({
    'City': ['A', 'B', 'C', 'D'],
    'Country': ['C1', 'C2', 'C3', 'C4'],
    'Latitude': [34.0522, 40.7128, 41.8781, 37.7749],
    'Longitude': [-118.2437, -74.0060, -87.6298, -122.4194]
}, geometry=gpd.points_from_xy(
    [-118.2437, -74.0060, -87.6298, -122.4194],
    [34.0522, 40.7128, 41.8781, 37.7749]
))

print(df)
df.plot(color='red', markersize=50)
plt.title('Geospatial Data Visualization')
plt.show()

m = folium.Map(
    location=[df.Latitude.mean(), df.Longitude.mean()],
    zoom_start=4
)
for _, r in df.iterrows():
    folium.Marker(
        [r.Latitude, r.Longitude],
        popup=f"{r.City}, {r.Country}"
    ).add_to(m)

m.save('geospatial_map.html')
m
```

How to Run

Save each code section in a separate .py file. Keep healthcare_dataset.csv in the same folder as the healthcare script. Run each file with Python. The geospatial script also saves an interactive map as geospatial_map.html.